**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 133 · Lesson 133 · Hashing for Integrity, Passwords and Data Lookup

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. How does hybrid encryption usually protect bulk web traffic?

Q2. What are the outputs of a half adder when A=1 and B=1?

Q3. In Big O notation, what is binary search worst-case complexity?

Q4. When does a linear search stop?

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# I Do · Read new knowledge

### Key vocabulary

hash function — Maps input data to a fixed-size or bounded-size result.

digest — Output of a cryptographic hash function.

collision — Two distinct inputs produce the same hash result.

salt — Random value combined with a password before secure password hashing.

### New learning

Compute a digest — A cryptographic hash produces a reproducible fingerprint of file content.

Compare for integrity — A matching trusted digest makes accidental/tamper changes less likely.

Choose correct use — Password hashing, integrity checking and hash-table indexing are different applications.

### Worked example

A publisher provides a trusted SHA-256 digest of a file. Explain how a pupil can check the download.

1. Trusted published digest — Find the given data and name the required result.

2. Calculate downloaded hash — Explain why this intermediate stage is necessary.

3. Compare exact digests — Calculate SHA-256 digest of received file and compare with the digest from a trusted source. If different, file changed or was corrupted. If equal, this strongly supports matching content, assuming trusted digest and secure algorithm; it does not by itself prove publisher identity.

# We Do · Practise with guidance

A database stores plaintext passwords. Explain a better use of hashing.

Choose random salt

Find the given data and name the required result.

Derive slow password hash

Explain why this intermediate stage is necessary.

Store salt plus result

Store unique salts and outputs from a slow password-hashing scheme such as Argon2id, not reversible encryption or a plain fast hash. On login, recompute using

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stored salt and compare securely.

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## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

State three different uses of hashing with one correct explanation each. [6]

### Question 2

Explain what a hash collision is and why it matters. [6]

### Question 3

Evaluate the claim 'a hash can decrypt a password and proves a file came from its claimed author'. [8]

## Exit Ticket · Answer before checking

A hash output can normally be decrypted to recover the exact original password.

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# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Use asymmetric establishment/authentication then symmetric traffic keys.

Q2. Sum 0, carry 1.

Q3. O(log n) for sorted random-access data.

Q4. On a match or after examining all items.

### You Do pages · Independent questions

Q1. File integrity checking [2], password verification [2], hash table key indexing/fast lookup [2].

Q2. Distinct inputs can share a hash [2], finite output range implies possible collisions [2], robust use selects collision-resistant function/handles table collisions [2].

Q3. Hashing is one way not decryption [2], passwords use salting and suitable password KDF [2], digest comparison aids integrity [2], origin needs trusted channel/signature [2].

### Exit Ticket · Model answer

False. Hashing is not reversible encryption.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

| Do Now | I Do | We Do | You Do | Review & Improve |
|---|---|---|---|---|
| Recall previous learning | Read new knowledge | Practise with guidance | Work independently | Check my understanding |